

A set A is strictly inductively ordered iff every nonempty totally ordered subset has a least upper bound. Assume that this set A is also nonempty, and partially ordered by $\leq$.

A map $f : A \rightarrow A$ is increasing if $\forall x \in A, x \leq f(x)$.

Let $a \in A$ and $B \subseteq A$ such that $a \in B$. B is an admissible set if $f(B) \subseteq B$ and for all non-empty totally ordered subsets T in A lies in B . This implies that B must also be strictly inductively ordered by definition.

Bourbaki-Witt Theorem – If A is a nonempty strictly inductively ordered and partially ordered set with $f : A \rightarrow A$ an increasing mapping, then $\exists x_0 \in A$ such that $f(x_0) = x_0$.

Proof. Let M be the intersection of all admissible subsets of A , where for the least upper bound $a \in A$, $a \in M$. Thus, M is nonempty. Now let $x \in M$, so then $f(x) \in M$. Thus, $f(M) \subseteq M$.

Let T be a totally ordered subset of M that is nonempty. Note that as $f(a) \in M$ then $f \circ f(a) \in M, \dots, f^{(n)}(a) \in M$.

Let c be an extreme point of M , meaning that when $x \in M$ and $x \leq c$, then $f(x) \leq c$. Let $M_c = \{x \in M : x \leq c \text{ or } f(c) \leq x\}$.

Lemma A – $M_c = M$ for every extreme point c of M .

This is identical to proving that M_c is an admissible set, which we shall do.

Proof. Let $x \in M_c$, and if $x < c$ and $f(x) \leq c$, then $f(x) \in M_c$. If $x = c$, $f(x) = f(c) \in M_c$.

If $f(c) \leq c$, then $f(c) \leq x \leq f(x)$ and so $f(M_c) \subseteq M_c$.

Let T be a nonempty totally ordered subset of M_c , and let b be the least upper bound of T in M . If $\forall x \in T, x \leq c$, then $b \leq c$ and $b \in M$.

If $\exists x \in T$ such that $f(c) \leq x$, then $f(c) \leq x \leq b$, so again $b \in M_c$. Thus, M_c is an admissible set, and so $M_c = M$. ■

Lemma B – Every element of M is an extreme point.

Proof. Let E be the set of all extreme points of M . E cannot be empty, since $a \in E$.

Let $c \in E$ and $x \in M$ such that $x < f(c)$. By Lemma A, $M = M_c$, so $x < c$ or $x = c$. $f(c) > x$ since $x < f(c)$.

If $x < c$, then $f(x) \leq c \leq f(c)$. If $x = c$, then $f(x) = f(c)$, so $f(E) \subseteq E$.

Let T be a nonempty totally ordered subset of E . Let b be the least upper bound of T in M .

Let $x \in M$ and assume $x < b$. Suppose $\exists c \in T$ such that $f(c) \leq x$, then $c \leq f(c) \leq x$ implies that x is an upper bound of T , so $b \leq x$, but this is not possible as $x < b$.

From Lemma A, $x \leq c$ for some $c \in T$. If $x < c$, then $f(x) \leq c \leq b$, and if $x = c$, then $c = x < b$. So, $f(x) \leq b$. Thus, $b \in E$ and E is admissible, so $E = M$. ■

Let $x, y \in M$, then by Lemma B, x is an extreme point of M , and suppose $y \in M_x$. This means that $y \leq x$ or $x \leq f(x) \leq y$, so M is totally ordered.

Then, there must exist a least upper bound $b \in M$ so that $b \leq f(b) \leq b$, so $f(b) = b$. ■

Zorn's Lemma – If S is a nonempty inductively ordered and partially ordered set, then there exists a maximal element in S .

Proof. Let A be the set of all nonempty totally ordered subsets of S . A is nonempty since A contains subsets of S with at least one element.

Let $X, Y \in A$ and let $X \leq Y$ means that $X \subseteq Y$. Then A is partially ordered by $\leq$ and is strictly inductively ordered since every nonempty totally ordered subset of A has a least upper bound, which is the union of all the sets in the subset.

Let $T = \{X_i\}_{i \in I}$ be a totally ordered subset of A , and let $Z = \bigcup_{i \in I} X_i$, which must be totally ordered since for $x, y \in Z$ then $x \in X_i$ and $y \in X_j$ for some $i, j \in I$, and as X_j is totally ordered, then either $x \leq y$ or $y \leq x$.

Suppose A does not have a maximal element. Then $\forall \Omega \in A, \exists \Omega' \in A$ such that $\Omega < \Omega'$. Let $f : A \rightarrow A$ be defined by $f(\Omega) = \Omega'$ for all $\Omega \in A$.

By the Bourbaki-Witt Theorem, $\exists \Omega_0 \in A$ such that $f(\Omega_0) = \Omega_0$, which is a contradiction.

Let X_0 be the maximal element of A .

Let m be an upper bound of X_0 , then m is the desired maximal element of S . If $x \in S$ and $m \leq x$, note then that $X_0 \cup \{x\}$ is totally ordered, which contradicts the maximality of X_0 . Thus, m is the maximal element of S . ■